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Indian Institute of Technology Hyderabad

Stochastic Processes

Lecture 26

Karthik P. N.
Assistant Professor, Department of AI
Email: pnkarthik@ai.iith.ac.in

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Upcrossing Lemma

Lemma (Upcrossing Lemma)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ *is a submartingale, then*

$$(b-a)\,\mathbb{E}[U_n] \;\leq\; \mathbb{E}[(X_n-a)_+] \;-\; \mathbb{E}[(X_0-a)_+] \qquad \forall\, n.$$

Remarks:

- Lemma 1 gives a control on the number of upcrossings
- If $\mathbb{E}[(X_n-a)_+] \;-\; \mathbb{E}[(X_0-a)_+]$ is not too large, then (intuitively):

small no. of upcrossings $\implies$ small number of oscillations $\implies$ almost-sure convergence.

Martingale Convergence Theorem

Theorem (Martingale Convergence Theorem)

If $\mathbf{X} = \{X_n\}_{n \geq 0}$ *is a submartingale with*

$$\sup_{n \geq 0} \mathbb{E}[(X_n)_+] \ < \ +\infty,$$

then $\mathbf{X}$ *converges almost surely. Furthermore, if the limit RV is denoted as* X*, then* $\mathbb{E}[|X|] < +\infty$.

Proof:

- Fix $a, b \in \mathbb{Q}$, $a < b$, and let $U_n[a, b]$ denote the number of upcrossings of $\mathbf{X}$ from a to b
- Clearly, $\{U_n[a, b]\}_{n \geq 0}$ is a non-negative, monotone increasing sequence, i.e.,

$$0 \ = \ U_0[a, b] \ \leq U_1[a, b] \ \leq \ U_2[a, b] \ \leq \ \cdots, \qquad U_n[a, b] \ \xrightarrow{\text{a.s.}} \ U[a, b].$$

  Here, $U[a, b]$ is the total number of upcrossings of $\mathbf{X}$ from a to b (measured until eternity)
- By MCT, it follows that

$$\lim_{n \to \infty} \mathbb{E}\Big[U_n[a, b]\Big] \ = \ \mathbb{E}\Big[U[a, b]\Big].$$

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Proof:

- Using the relation $(Z-a)_+ \leq Z_+ + |a|$, it follows from Lemma 1 that

$$\mathbb{E}[U_n[a,b]] \ \leq \ \frac{\mathbb{E}[(X_n-a)_+] \ - \ \mathbb{E}[(X_0-a)_+]}{b-a} \ \leq \ \frac{\mathbb{E}[(X_n-a)_+]}{b-a} \ \leq \ \frac{\mathbb{E}[(X_n)_+] + |a|}{b-a}$$

- Taking limits as $n \to \infty$ on both sides and noting that $\{U_n[a,b]\}_n$ is monotone, we get

$$\mathbb{E}[U[a,b]] \ = \ \lim_{n \to \infty} \mathbb{E}[U_n[a,b]] \ = \ \sup_n \mathbb{E}[U_n[a,b]] \ \leq \ \frac{\sup_n \mathbb{E}[(X_n)_+] + |a|}{b-a} \ < \ +\infty.$$

Martingale Convergence Theorem

Theorem (Martingale Convergence Theorem)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ *is a submartingale with* $\sup_{n\geq 0} \mathbb{E}[(X_n)_+] < +\infty,$ *then* $\mathbf{X}$ *converges almost surely. Furthermore, if the limit RV is denoted as* X*, then* $\mathbb{E}[|X|] < +\infty$.

Proof:

- Because $U[a,b]$ is a non-negative RV, we have

$$\mathbb{E}[U[a,b]] < +\infty \implies \mathbb{P}(\{U[a,b] < +\infty\}) = 1.$$

- This means that

$$\mathbb{P}\left(\left\{\liminf_{n\to\infty} X_n < a < b < \limsup_{n\to\infty} X_n\right\}\right) = 0.$$

- Because the above holds for all rational $a < b$, we get

$$\mathbb{P}\left(\underbrace{\left\{\liminf_{n\to\infty} X_n = \limsup_{n\to\infty} X_n\right\}}_{A}\right) = 1.$$

Martingale Convergence Theorem

Theorem (Martingale Convergence Theorem)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ is a submartingale with $\sup_{n\geq 0} \mathbb{E}[(X_n)_+] < +\infty$, then $\mathbf{X}$ converges almost surely. Furthermore, if the limit RV is denoted as X, then $\mathbb{E}[|X|] < +\infty$.

Proof:

- Let X denote the limit RV
  (defined to be the common value of $\liminf$ and $\limsup$ on A, and 0 on $A^{\complement}$)
- We then have

$$\mathbb{E}[X_+] = \mathbb{E}\left[\lim_{n\to\infty}(X_n)_+\right] = \mathbb{E}\left[\liminf_{n\to\infty}(X_n)_+\right] \leq \liminf_{n\to\infty} \mathbb{E}[(X_n)_+] = \sup_n \mathbb{E}[(X_n)_+] < +\infty.$$

- We also have

$$\begin{aligned}\mathbb{E}[X_-] = \mathbb{E}\left[\liminf_{n\to\infty}(X_n)_-\right] &\leq \liminf_{n\to\infty} \mathbb{E}[(X_n)_-] \leq \liminf_{n\to\infty} \mathbb{E}[(X_n)_+] - \mathbb{E}[X_n] \\ &\leq \liminf_{n\to\infty} \mathbb{E}[(X_n)_+] - \mathbb{E}[X_0] = \sup_{n\to\infty} \mathbb{E}[(X_n)_+] - \mathbb{E}[X_0] < +\infty.\end{aligned}$$

A Corollary to Theorem 2

- **A non-negative supermartingale $\mathbf{X} = \{X_n\}_{n \geq 0}$ converges a.s.**
  The limit RV, say X, satisfies $\mathbb{E}[X] \leq \mathbb{E}[X_0]$
  The result follows from applying Theorem 2 to $Y_n = -X_n$ and noting that $\sup_n \mathbb{E}[(Y_n)_+] = 0$

Martingale with Bounded Increments

Theorem (Martingale with Bounded Increments)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ is a **martingale with bounded increments** *(i.e., $|X_n - X_{n-1}| \leq M$ for some $0 < M < +\infty$), then one of the following two events occurs with probability 1:*

$$C = \{\lim_{n\to\infty} X_n \text{ exists and is finite}\}, \qquad D = \{\limsup_{n\to\infty} X_n = +\infty, \quad \liminf_{n\to\infty} X_n = -\infty\}.$$

Proof of Theorem 6:

- Fix $K \in \mathbb{N}$. Let $\tau_K := \inf\{n : X_n \leq -K\}$
- We know that $\{X_{\tau_K \wedge n}\}_{n\geq 0}$ is a martingale
- Note that

$$X_{\tau_K} \leq -K, \qquad X_{\tau_K - 1} > -K, \qquad X_{\tau_K} = X_{\tau_K} - X_{\tau_K - 1} + X_{\tau_K - 1} \geq -M - K.$$

- If $n < \tau_K$, then $X_{\tau_K \wedge n} > -K \geq -K - M$
- Combining, we can say that $\{X_{\tau_K \wedge n} + K + M\}_{n\geq 0}$ is a non-negative martingale
  By Theorem 2, we can say that $\{X_{\tau_K \wedge n} + K + M\}_{n\geq 0}$ converges a.s. to a limit RV that is finite a.s.

Martingale with Bounded Increments

Theorem (Martingale with Bounded Increments)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ is a **martingale with bounded increments** *(i.e., $|X_n - X_{n-1}| \leq M$ for some $0 < M < +\infty$), then one of the following two events occurs with probability 1:*

$$C = \{\lim_{n\to\infty} X_n \text{ exists and is finite}\}, \qquad D = \{\limsup_{n\to\infty} X_n = +\infty, \quad \liminf_{n\to\infty} X_n = -\infty\}.$$

Proof of Theorem 6:

- Observe that

$$\forall\, K \in \mathbb{N}, \quad \{\tau_K = +\infty\} \subseteq C, \qquad \{\liminf_{n\to\infty} X_n > -\infty\} \subseteq \bigcup_{K\in\mathbb{N}} \{\tau_K = +\infty\} \subseteq C.$$

- Applying all previous arguments to the bounded martingale $Y_n = -X_n$, we get that

$$\{\limsup_{n\to\infty} X_n < +\infty\} = \{\liminf_{n\to\infty} -(X_n) > -\infty\} \subseteq C.$$

- Thus, it follows that $C^{\complement} \subseteq D$

Doob's Decomposition

Lemma (Doob's Decomposition)

If $\mathbf{X} = \{X_n\}_{n\geq 0}$ *is a submartingale, then it can be decomposed as*

$$X_n = M_n + A_n,$$

where $\mathbf{M} = \{M_n\}_{n\geq 0}$ *is a* **martingale**, *and* $\mathbf{A} = \{A_n\}_{n\geq 0}$ *is a* **predictable, non-decreasing sequence** *with* $A_0 = 0$.

Proof:

- We want:

$$X_n = M_n + A_n, \qquad \mathbb{E}[M_n \mid \mathscr{G}_{n-1}] = M_{n-1}, \qquad A_n \text{ is } \mathscr{G}_{n-1}\text{-measurable}$$

- The proof involves working backwards to construct $\mathbf{M}$ and $\mathbf{A}$ satisfying the above properties
- We must have

$$\mathbb{E}[X_n \mid \mathscr{G}_{n-1}] = M_{n-1} + A_n = X_{n-1} - A_{n-1} + A_n,$$

  from which it follows that

$$A_n - A_{n-1} = \mathbb{E}[X_n \mid \mathscr{G}_{n-1}] - X_{n-1} = \mathbb{E}[X_n - X_{n-1} \mid \mathscr{G}_{n-1}] \qquad \forall\, n.$$

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where $\mathbf{M} = \{M_n\}_{n \geq 0}$ *is a* **martingale**, *and* $\mathbf{A} = \{A_n\}_{n \geq 0}$ *is a* **predictable, non-decreasing sequence** *with* $A_0 = 0$.

Proof:

- Defining $A_0 = 0$, we then construct $\mathbf{A}$ as

$$A_n = \sum_{k=1}^{n} \mathbb{E}[X_k - X_{k-1} \mid \mathscr{G}_{k-1}], \qquad n \in \mathbb{N}.$$

- Defining $M_0 = X_0$, we define $\mathbf{M}$ via

$$M_n = X_n - A_n, \qquad n \in \mathbb{N}.$$

- It is easy to verify that $\mathbf{A}$ is a predictable, non-decreasing sequence, and that $\mathbf{M}$ is a martingale

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Ville's Inequality

Lemma (Ville's Inequality)

Let $\mathbf{X} = \{X_n\}_{n \geq 0}$ *be a* **non-negative supermartingale**. *Then, for all* $\varepsilon > 0$,

$$\mathbb{P}\left(\left\{\exists\, n \geq 0 : \; X_n \geq \varepsilon\right\}\right) \;\leq\; \frac{\mathbb{E}[X_0]}{\varepsilon}.$$

Proof:

- Fix $\varepsilon > 0$, and let τ_ε be defined as

$$\tau_\varepsilon \;:=\; \inf\{n \geq 0 : \; X_n \geq \varepsilon\}.$$

- By Doob's optional stopping theorem, we have

$$\mathbb{E}[X_0] \;\geq\; \mathbb{E}[X_{\tau_\varepsilon} \mathbf{1}_{\{\tau_\varepsilon < +\infty\}}] \;\geq\; \varepsilon\, \mathbb{P}(\{\tau_\varepsilon < +\infty\}).$$

  Rearranging terms, we arrive at the desired inequality